

$$\mathcal{P}(\mathbb{R}) = \{ \{1\}, \{2\}, \{-1\}, \{3\}, \dots \\ \{1,2\}, \{1,3\}, \{1,4\}, \dots \\ \{-1,1\}, \{-1,2\}, \dots \\ \{20.1\}, \{20.2\}, \{20.3\}, \dots \}$$

$\mathbb{R}^2 = \{ \text{cartesian plane} \}$

$$\mathbb{R}^2 = \{ (1,1), (1.1,2), (1.3,1), (1.4,1), \dots \}$$

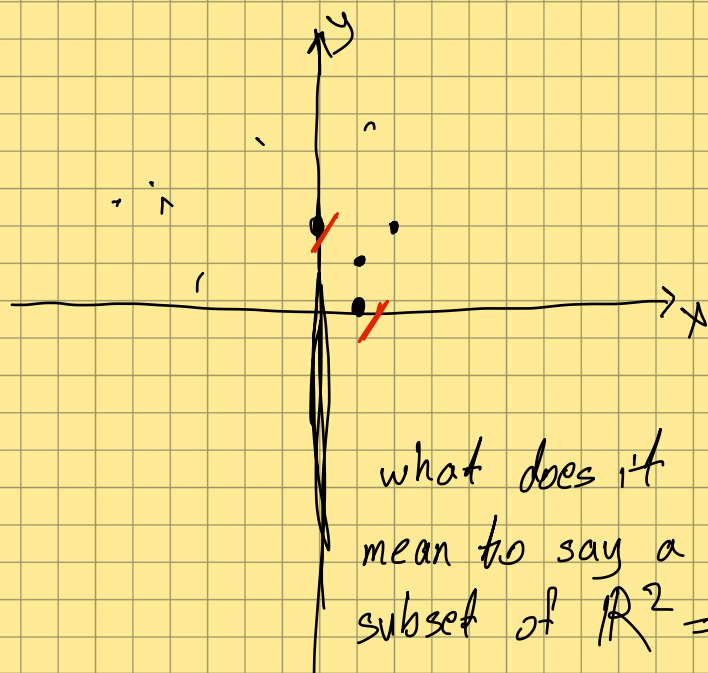
$\vdots$

$\{$

$$\mathcal{P}(\mathbb{R}^2) = \{ (x,y) : x,y \in \mathbb{R} \} = \mathbb{R}^2$$

infinite

$$\mathcal{P}(\mathbb{R}^2) = \{ X : X \subseteq \mathbb{R}^2 \}$$



$$\begin{aligned} \{ (1,1) \} &\subseteq \mathbb{R}^2 \checkmark \\ \{ (1,1), (2,2) \} &\subseteq \mathbb{R}^2 \checkmark \\ \{ (1,1), (2,2), (0,2) \} &\subseteq \mathbb{R}^2 \checkmark \\ \{ (1,0), (1,1), (2,2), (0,2) \} &\subseteq \mathbb{R}^2 \checkmark \\ \{ (0,2), (1,0) \} &\subseteq \mathbb{R}^2 \end{aligned}$$